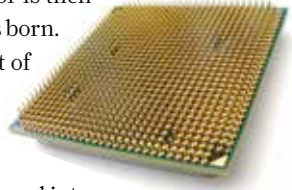


that creep into certain processors. These processors are later tested and a stable frequency is achieved. The processor is then locked onto this frequency and a new SKU is born. All features that are not supposed to be part of that SKU are burned out with a laser. Naturally, the high end processors were the best performing from a particular bunch, that is why they tested well and subsequently got binned into a higher category. Enthusiasts also like to bin processors, this helps figure out which batch had the best scaling processors and then that particular batch sells like hot-cakes.



7. Manufacturing process

Processors are made up of billions of transistors and they are connected by thin wiring, it is the width of the wiring which denotes the manufacturing process. The smaller the manufacturing process the more transistors that can be squeezed into a small area and thus better performance can be achieved. Both AMD and Intel outsource their manufacturing to the same company and they share similar trends when shifting between manufacturing processes. Intel calls its trend as Tick-Tock. The Tick refers to a reduction in manufacturing process (smaller transistor size) while the Tock refers to a new architecture revision. They occur in an alternate fashion.

8. Cache

Cache is the shared memory between processors. There are different levels of cache depending upon the processor architecture. Each core has its own L1 and L2 cache. All the cores are then linked to each other via the L3 cache. Since the L1 is the closest to any core, not much memory is needed (data is stored and retrieved really fast) but as we move further away and get to the L3 cache we need a lot more. The basic idea is that more is better.

9. TDP

Thermal Design Process refers to the net wattage that a processor will consume. Greater the TDP the more heat it will dissipate and thus you need a better heatsink. Better performing processors naturally have more TDP but we are seeing a gradual reduction in TDP each processor generation since the manufacturing process is getting smaller and smaller. There are certain variants of high-end processors that come on low TDP SKUs. These